

[Total No. of Questions - 9] [Total No. of Printed Pages - 3]
Dec-25-0547

MCA-7301 (Operational Research)

MCA-3rd NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

SECTION-A

1. Explain the mathematical formulation of linear programming problems (LPP) in detail. Discuss the general form of an LPP with constraints, objective function, and assumptions. Illustrate using a suitable example from manufacturing or resource allocation. (12)
2. A company manufactures two products A and B. Each unit of product A requires 3 hours on machine I and 2 hours on machine II. Each unit of product B requires 2 hours on machine I and 4 hours on machine II. The maximum time available on machine I is 120 hours and on machine II is 160 hours per week. The profit per unit for product A is ₹50 and for product B is ₹60. Formulate this as a linear programming problem and solve it using the graphical method to determine the optimal production plan. (12)

SECTION - B

3. Discuss the assignment problem in detail. Explain the Hungarian method for solving assignment problems. How do you handle unbalanced assignment problems and problems

2.

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involving maximization? Explain using a suitable and complete numerical example. (12)

4. Describe project scheduling using CPM (Critical Path Method) and PERT (Program Evaluation and Review Technique). Explain network construction, forward and backward pass calculations, determination of critical path, and float calculations. What are the key differences between CPM and PERT? (12)

SECTION-C

5. Describe the maximin-minimax principle in game theory. How do you determine the value of a game? Explain the concept of saddle point and pure strategy solutions. Provide an example of games with and without saddle points. (12)
6. A bank has one teller. Customers arrive at an average rate of 20 per hour according to Poisson distribution. The teller can serve customers at an average rate of 25 per hour following exponential distribution. Calculate: (a) Average number of customers in the system, (b) Average waiting time in queue, (c) Probability that the teller is idle, (d) Probability of having more than 3 customers in the system. (12)

SECTION - D

7. Discuss the EOQ model with shortage (planned backorders). How does it differ from the basic EOQ model? Derive the formulas for optimal order quantity and maximum shortage level. When is this model applicable in practice? (12)
8. Describe inventory control techniques for uncertain demand. Explain the concept of safety stock, reorder point, and service level. How do you handle stochastic demand using probabilistic inventory models? Discuss the single-period inventory problem. (12)

[P.T.O.]

SECTION - E (Compulsory)

9. (a) What are the exceptional cases that can occur while solving linear programming problems? (2)
- (b) What is crashing in project management? (2)
- (c) Define slack time and float in project scheduling. (2)
- (d) Define the following queueing theory terms: service rate and queue discipline. (2)
- (e) Define lead time and safety stock. (2)
- (f) Briefly explain the trade-offs between ordering costs, holding costs, and shortage costs. (2)